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FEM Solver Behavior Report on a Femur Bone: Linear vs. Neo-Hookean Material Models

In this simulation study, we compare the behavior of the finite element method (FEM) solver for two different material models applied to a femur bone geometry: a linear isotropic elasticity model and a nonlinear Neo-Hookean hyperelastic model. Both simulations were executed using identical meshes and boundary conditions.

The linear model was defined using an isotropic elasticity formulation with a Young's modulus of 1.7×10^{10} Pa and a Poisson's ratio of 0.3, resulting in a shear modulus of 6.5385×10^9 Pa. This model assumes small deformations and linear stress-strain behavior, making it suitable for efficient, direct computation. It required only one iteration, with no need for time stepping or convergence refinement.

In contrast, the nonlinear model was constructed using a Neo-Hookean hyperelastic material with the same shear modulus and an incompressibility parameter D1 of 1.4188×10^{-10} Pa⁻¹. Large deflections were enabled to support nonlinear geometric effects. The solver used the full Newton-Raphson method with automatic time stepping, resulting in six substeps and a total of 30 equilibrium iterations. Each substep required multiple iterations to reduce both force and displacement residuals below tolerance. At substep 4, the solver encountered a divergence in the solution and responded by halving the time increment to recover convergence.

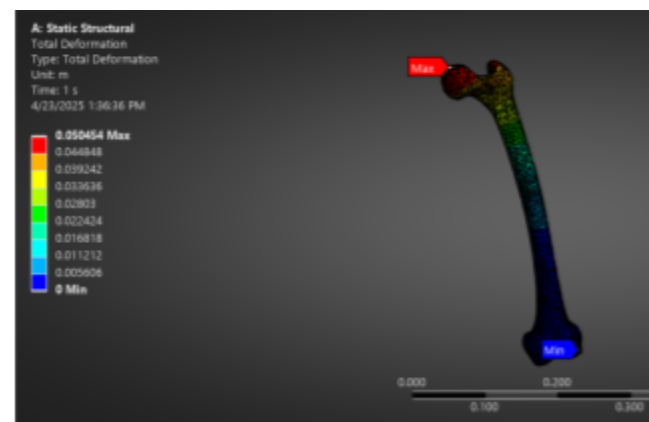


Figure 1: linear model's Total deformation

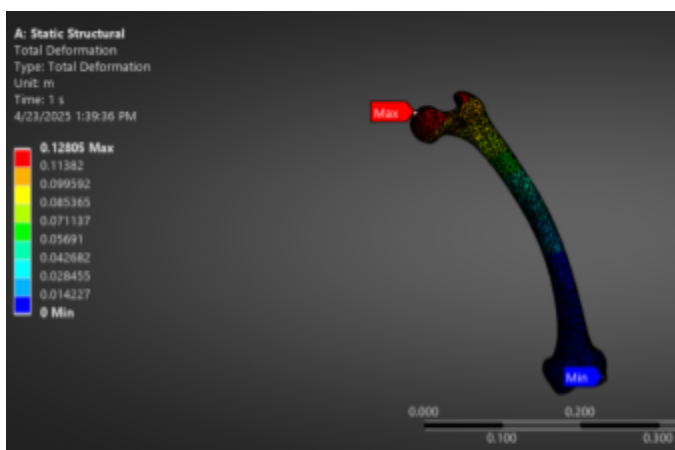


Figure 2: Neo-Hookean model's total deformation

While the nonlinear simulation introduced significantly higher computational costs (~62 seconds vs. 1 second for the linear model), it provided a more realistic depiction of large-strain material behavior. The linear model, by contrast, was efficient and numerically stable but limited in its ability to capture complex deformation responses. These results emphasize that while linear models are appropriate for small deformations, nonlinear models like Neo-Hookean formulations are essential for accurate representation under large loads.